Week 08: October 19-23
C 3.2 Defence against disease

Subject/:الموضوع Biology **Gradeالصف : 12DP**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1	I can explain how the body prevents and controls infection by relating pathogens, physical barriers, blood clotting, innate immunity and phagocytosis to the body's early defence against infectious disease.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 2	I can explain how adaptive immunity produces a specific and lasting response by tracing antigen recognition, helper T-cell activation of B-lymphocytes, clonal expansion, antibody secretion and memory-cell formation.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 3	I can analyze how infectious diseases are transmitted, treated and evolve by explaining HIV infection, selective antibiotic action, evolution of antibiotic resistance and transmission of zoonotic pathogens.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 4	I can evaluate how vaccination contributes to individual and population-level protection by explaining immunization and herd immunity	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models.	IB Biology, InThinking

	and analyzing epidemiological data from the COVID-19 pandemic.	7.2 Make learning relevant, authentic, and connected to students' backgrounds.	
Homework الواجبات			
Teacher's notes/ Reminders ملاحظات المعلم والتذكير			

Week # 08 الأسبوع#

Technology Integration دمج التكنولوجيا	**digital microscopy and high-resolution cell imagery** to examine phagocytes and relate cell morphology to immune function **animated phagocytosis sequence** to visualize recognition, engulfment, vesicle formation and intracellular digestion model of skin and mucous membranes to examine how physical barriers reduce pathogen entry wound formation and clotting digitally to demonstrate how disruption and restoration of a physical barrier influence infection risk **immune-response model** to visualize antigen recognition and selection of specific lymphocyte populations molecular visualizations to demonstrate why antibody–antigen interactions are specific animated clonal-expansion models to follow one activated B cell as it generates plasma cells and memory cells digital graphing to compare simulated primary and secondary antibody responses
Cross-curricular links **الروابط بين المواد الدراسية**	**Mathematics** • Percentage change • Rates • Ratios • Graph interpretation • Statistical comparison **Health Sciences** • Vaccination • Epidemiology • Population health • Disease prevention Chemistry • Selective chemical inhibition • Drug-target interactions Mathematics • Percentage resistance • Rate of change • Trend analysis Health Sciences • Antimicrobial resistance • HIV prevention

	• Infectious-disease management Environmental Science • Human–animal interactions • Zoonotic disease emergence **Chemistry** • Protein structure and function • Enzymatic digestion within lysosomes • Fibrin formation **Health Sciences** • Infection prevention • Wound care • Disease transmission **Mathematics** • Interpreting infection-frequency data • Comparing rates and proportions
External Assessment Integration دمج التقييمات الخارجية	**Paper 1** MCQs on: • HIV transmission • HIV and lymphocytes • Antibiotic specificity • Antibiotic resistance • Natural selection • Zoonoses **Paper 2** Practice: • Interpret antibiotic-resistance graphs. • Explain resistance using natural selection. • Distinguish HIV from AIDS. • Analyze infectious-disease transmission scenarios. • Explain why antibiotics have selective effects on bacterial cells.

	Paper 1 MCQs on: • Antigens • B-lymphocytes • Helper T-lymphocytes • Plasma cells • Memory cells • Antibody specificity Paper 2 Practice: • Explain activation of B cells. • Explain clonal expansion. • Interpret antibody-concentration graphs. • Explain immunological memory. • Apply adaptive-immunity mechanisms to novel scenarios. Paper 1 MCQs on: • Pathogen identification • Primary defences • Blood clotting • Innate vs adaptive immunity • Phagocytosis Paper 2 Practice: • Explain how skin prevents infection. • Explain how clotting contributes to defence. • Compare innate and adaptive immunity. • Annotate a phagocytosis sequence. • Apply immune-defence concepts to unfamiliar infection scenarios.
Wellbeing/ ATL Skills **الصحة النفسية/ أساليب التعلم**	**Research & Thinking (ATL)** • Classifying defence mechanisms.

	• Linking structure to protective function. • Analyzing cause-and-effect relationships during infection. • Applying mechanisms to unfamiliar clinical scenarios. **Communication** • Explaining defence pathways sequentially. • Using precise immunological vocabulary. • Constructing annotated biological diagrams. **Collaboration** • Developing group defence models. • Comparing and refining explanations using biological evidence.
Standards **المعايير**	C3.2.1 Pathogens as the cause of infectious diseases C3.2.2 Skin and mucous membranes as a primary defence C3.2.3 Sealing of cuts in skin by blood clotting C3.2.4 Differences between the innate immune system and the adaptive immune system C3.2.5 Infection control by phagocytes C3.2.6 Lymphocytes as cells in the adaptive immune system that cooperate to produce antibodies C3.2.7 Antigens as recognition molecules that trigger antibody production C3.2.8 Activation of B-lymphocytes by helper T-lymphocytes C3.2.9 Multiplication of activated B-lymphocytes to form clones of antibody-secreting plasma cells C3.2.10 Immunity as a consequence of retaining memory cells C3.2.11 Transmission of HIV in body fluids C3.2.12 Infection of lymphocytes by HIV with AIDS as a consequence C3.2.13 Antibiotics as chemicals that block processes occurring in bacteria but not in eukaryotic cells C3.2.14 Evolution of resistance to several antibiotics in strains of pathogenic bacteria C3.2.15 Zoonoses as infectious diseases that can transfer from other species to humans C3.2.16 Vaccines and immunization C3.2.17 Herd immunity and the prevention of epidemics C3.2.18 Evaluation of data related to the COVID-19 pandemic
Inquiry/ TOK Question **البحث والاستقصاء أسئلة TOK**	How does scientific classification help us understand a complex system when its components, such as innate and adaptive immunity, interact continuously?

	To what extent does describing the immune system as having “memory” help or potentially mislead our understanding of the biological mechanism?
Success Criteria معايير النجاح (Measurable)	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: Lesson 1 MUST **– Recall:** I can **identify and describe** pathogens, primary barriers, blood clotting, phagocytes, and the major differences between innate and adaptive immunity. SHOULD **– Apply & Analyze:** I can **explain how skin, mucous membranes, clotting and phagocytosis work together to prevent or control infection.** COULD **– Evaluate & Create:** I can **construct and justify an integrated defence model predicting how disruption of one primary or innate defence could influence the progression of infection.** Lesson 2: MUST **– Recall:** I can **identify and describe** antigens, helper T-lymphocytes, B-lymphocytes, plasma cells, antibodies and memory cells. SHOULD **– Apply & Analyze:** I can **trace and explain the adaptive immune response from antigen recognition through B-cell activation, clonal expansion, antibody production and memory-cell formation.** COULD **– Evaluate & Create:** I can **analyze immune-response data and construct an evidence-based explanation of how cellular interactions generate specificity and long-term immunity.** Lesson 3: MUST **– Recall:** I can **identify and describe** HIV transmission, AIDS, antibiotic action, antibiotic resistance and zoonoses. SHOULD **– Apply & Analyze:** I can **explain HIV's effects on adaptive immunity and apply natural selection to explain the evolution of antibiotic-resistant bacterial populations.** COULD **– Evaluate & Create:** I can **analyze infectious-disease and resistance data to construct an evidence-based explanation linking pathogen biology, evolution, treatment and transmission.** Lesson 4: MUST **– Recall:** I can **identify and describe** vaccination, immunization, memory cells, herd immunity, epidemics and key epidemiological measures. SHOULD **– Apply & Analyze:** I can **explain how vaccination contributes to individual and population protection and analyze epidemiological data using rates, trends and biological reasoning.** COULD **– Evaluate & Create:** I can **evaluate COVID-19 epidemiological evidence by considering confounding variables, limitations, correlation and causation and construct a justified conclusion supported by multiple pieces of evidence.**

National Identity Domain **مجالات الربط بالهوية الوطنية**	Choose an item.	Global Understanding التفاهم العالمي	Choose an item.
	المجال **Area:** Community health, infection prevention, personal responsibility and care for others **Area:** Immunity, public health, scientific literacy and shared responsibility for disease prevention		
	التطبيق **Implementation:** Students connect infection-control mechanisms with responsible behaviors that protect both individuals and the wider UAE community. Case discussions emphasize how hygiene, wound care and responsible responses to infectious disease demonstrate concern for the wellbeing of others in a diverse society. **Implementation:** Students consider how immune-system knowledge contributes to public health in the UAE's diverse population. Discussion emphasizes that infectious diseases cross geographical and social boundaries and that understanding immunity supports informed and respectful health decisions in multicultural communities.		